



ABENA ALEX NELSON RYAN

Junior Backend Engineer

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<https://ninjashadowboy.github.io/portfolio-app/> | <https://github.com/ninjashadowboy>

Languages: French (native), English (C1 - IELTS 7.5)

Profile

Backend-focused software engineering student with strong experience building scalable APIs, containerized systems, and a custom C++ ECS game engine. Skilled across Spring Boot, NestJS, FastAPI, and Docker, with a systems-level foundation gained through both professional roles and low-level C/C++ work. Applies AI integrations (RAG, LLM, sentiment analysis) to solve practical problems in African institutional contexts. Ranked **1st/68** in third year (GPA: **3.49/4**). Selected projects available on [GitHub](#).

Professional Experience

Junior Backend Developer (NestJS) at ITIAD

Sept 2025 - Feb 2026

- Led design and implementation of JWT-based authentication system for IMMOA real estate marketplace, securing 50+ API endpoints with 15-minute access / 7-day refresh token lifecycle
- Improved backend reliability by implementing comprehensive unit tests, achieving **70% code coverage** and identifying 12+ critical edge cases in payment workflows
- Automated API documentation using Swagger/OpenAPI, enabling faster frontend-backend parallel development

Intern at ORION

July - Aug 2025

- Explored hardware-software interactions through deep study of /dev, /proc, /sys, and Linux system calls
- Built a Kotlin command-line system monitor that parses kernel and device information in real time
- One of the first members of the new hardware systems department

Academic Intern at CENADI

June - Aug 2023

- Audited and documented physical infrastructure inventory (servers, workstations, network equipment) across the data center
- Analyzed network configuration and system architecture, producing a comparative report benchmarked against industry reference models

Research & Publications

Co-author: "Bilingual LLM Campus Assistant via RAG and LMS Integration"

Research Methodology Paper - Institut Universitaire Saint Jean du Cameroun (April 2026)

Link: <https://sites.google.com/view/ds4dhresearchlab/workshops/upcoming-workshop>

- Designed and implemented core AI chatbot, component integration, and cloud deployment (FastAPI RAG service with ChromaDB)
- Followed Nunamaker methodology with TAM evaluation involving 23 students (91% adoption intent, mean scores >4.2)
- Supervised by Dr. Philippe Tamla

Education

Computer and Information System Engineering

Institut Universitaire Saint Jean du Cameroun (2022 - Present)

GPA: 3.49/4, Ranked 1st out of 68 in third year

GCE Advanced Level with 23/25 points

Maths A, Further Maths A, ICT A, Physics B, Chemistry B

GBHS Mbalmayo (2020 - 2022)

Ranked 1st out of 68 in graduating class (average 17.99/20)

Technical Skills

Core Languages (Intermediate+):	Java, Kotlin, Python, C, C++
Backend & Architecture:	Spring Boot, NestJS, FastAPI, Microservices, REST APIs, JWT
AI/ML & Data:	RAG, LLM Integration (Gemini), ChromaDB, Embeddings, Sentiment Analysis
Infrastructure & DevOps:	AWS EC2, Docker & Docker Compose, GitLab CI/CD, PostgreSQL, MongoDB (Beginner)
Cybersecurity:	Cyber Kill Chain methodology, Metasploit (msfconsole), penetration testing concepts
Frontend & Mobile:	Angular, React Native (offline-first), TypeScript, Jetpack Compose
Other:	Algorithms & Data Structures, Software Testing, SDL Game Programming, Agile (Scrum)

Projects (Selected)

- AcadAi** (2025-2026) (*Capstone Project*)
LLM-powered bilingual campus assistant (RAG + Quizzzy)

 - Designed Python FastAPI RAG service with ChromaDB for bilingual institutional grounding
 - Deployed on AWS with Docker; led AI chatbot & component integration; supervised TAM evaluation (91% adoption intent)

Lora (2025-2026)
Offline-first agricultural platform (Cameroon)

 - Spring Boot microservices + React Native (offline-first)
 - Built offline-first React Native app with sync queue for intermittent connectivity
 - AWS EC2 deployment + plant health analysis

Quizzzy (2025-2026)
Learning management system with MCQ assessments

 - Built all REST API endpoints and database queries using NestJS; integrated Gemini LLM for sentiment analysis on student responses

- Developed Angular admin dashboard surfacing school-wide KPIs: exam statistics, attendance rates, and score distributions

MeetIn (2025-2026)
Enterprise room booking system

 - Designed and implemented a custom RBAC permission system from scratch, including Role entity modeling, permission assignment, and enforcement across all Spring Boot endpoints

C++ ECS Game Engine (SDL)

 - Modular 2D engine with Entity-Component-System architecture, scene management and physics components

Hotel Hub (2024)
Hotel reservation system (Spring Boot)

 - Stripe payments, authentication & email notifications

Soft Skills

Teamwork, Public Speaking, Teaching, Problem Solving

Hobbies

Basketball (Point Guard), Taekwondo (Yellow Belt), Chess (400 Elo)